

**THE THIRTY-THIRD WILLIAM LOWELL PUTNAM
MATHEMATICAL COMPETITION**

December 2, 1972

- A-1. Show that there are no four consecutive binomial coefficients $\binom{n}{r}, \binom{n}{r+1}, \binom{n}{r+2}, \binom{n}{r+3}$ (n, r integers > 0 and $r+3 \leq n$) which are in arithmetic progression.

- A-2. Let S be a set and let $*$ be a binary operation on S satisfying the laws

$$x * (x * y) = y \quad \text{for all } x, y \text{ in } S,$$

$$(y * x) * x = y \quad \text{for all } x, y \text{ in } S.$$

Show that $*$ is commutative but not necessarily associative.

- A-3. If for a sequence $x_1, x_2, x_3, \dots$, $\lim_{n \rightarrow \infty} (x_1 + x_2 + \dots + x_n)/n$ exists, call this limit the C -limit of the sequence. A function $f(x)$ from $[0, 1]$ to the reals is called a supercontinuous function on the interval $[0, 1]$ if the C -limit exists for the sequence $f(x_1), f(x_2), f(x_3), \dots$ whenever the C -limit exists for the sequence $x_1, x_2, x_3, \dots$. Find all supercontinuous functions on $[0, 1]$.

- A-4. Of all ellipses inscribed in a square, show that the circle has the maximum perimeter.

- A-5. Show that if n is an integer greater than 1, then n does not divide $2^n - 1$.

- A-6. Let $f(x)$ be an integrable function in $0 \leq x \leq 1$ and suppose $\int_0^1 f(x)dx = 0, \int_0^1 x f(x)dx = 0, \dots, \int_0^1 x^{n-1} f(x)dx = 0$ and $\int_0^1 x^n f(x)dx = 1$. Show that $|f(x)| \geq 2^n/(n+1)$ in a set of positive measure.

- B-1. Show that the power series representation for the series $\sum_{n=0}^{\infty} (x^n(x-1)^{2n})/n!$ cannot have three consecutive zero coefficients.

- B-2. A particle moving on a straight line starts from rest and attains a velocity v_0 after traversing a distance s_0 . If the motion is such that the acceleration was never increasing, find the maximum time for the traverse.

- B-3. Let A and B be two elements in a group such that $ABA = BA^2B, A^3 = 1$ and $B^{2n-1} = 1$ for some positive integer n . Prove $B = 1$.

- B-4. Let n be an integer greater than 1. Show that there exists a polynomial $P(x, y, z)$ with integral coefficients such that $x \equiv P(x^n, x^{n+1}, x + x^{n+2})$.

- B-5. If the opposite angles of a skew (non-planar) quadrilateral are equal in pairs, prove that the opposite sides are equal in pairs.

- B-6. Let $n_1 < n_2 < n_3 < \cdots < n_k$ be a set of positive integers. Prove that the polynomial $1 + z^{n_1} + z^{n_2} + \cdots + z^{n_k}$ has no roots inside the circle $|z| < (\sqrt{5} - 1)/2$.